

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



FLOWER MUSK

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	29.09.2023	300001120343	Date of first issue: 29.09.2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : FLOWER MUSK

Sales Number : 6217241

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub-stance/Mixture : Fragrance for consumer product

1.3 Details of the supplier of the safety data sheet

Company : IFF (NEDERLAND) BV
ZEVENHEUVELENWEG 60
5048 AN TILBURG

Telephone : +31134642211

Telefax : +31134636032

E-mail address of person responsible for the SDS : sds@iff.com

1.4 Emergency telephone number

ORFILA (INRS): + 33 (0)1 45 42 59 59

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Skin irritation, Category 2 H315: Causes skin irritation.

Eye irritation, Category 2 H319: Causes serious eye irritation.

Skin sensitisation, Category 1 H317: May cause an allergic skin reaction.

Long-term (chronic) aquatic hazard, Category 2 H411: Toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

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Hazard pictograms :



Signal word : Warning

Hazard statements :
H315 Causes skin irritation.
H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.
H411 Toxic to aquatic life with long lasting effects.

Precautionary statements : **Prevention:**
P261 Avoid breathing mist or vapours.
P264 Wash skin thoroughly after handling.
P280 Wear protective gloves/ eye protection/ face protection.
Response:
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P337 + P313 If eye irritation persists: Get medical advice/ attention.
P391 Collect spillage.

Hazardous components which must be listed on the label:

68155-66-8,	Tetramethyl Acetyloctahydronaphthalenes
54464-57-2,	
68155-67-9,	
54464-59-4	
78-70-6	linalool
33704-61-9	Dihydro pentamethylindanone
63286-42-0	OXACYCLOHEPTADEC-10-EN-2-ONE
5989-27-5	(R)-p-mentha-1,8-diene
104-54-1	CINNAMYL ALCOHOL
11031-45-1	santalol
107-75-5	HYDROXYCITRONEL LAL
127-91-3	BETA-PINENES
97-54-1	isoeugenol
23696-85-7	ROSE KETONE-4

2.3 Other hazards

None reasonably foreseeable.

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

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SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number		Concentration (% w/w)
1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one	68155-66-8 915-730-3 01-2119489989-04	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Chronic 2; H411	>= 10 - < 20
linalool	78-70-6 201-134-4 603-235-00-2 01-2119474016-42	Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1B; H317	>= 1 - < 10
1,4-dioxacyclohexadecane-5,16-dione	54982-83-1 259-423-6 01-2119524000-64	Aquatic Acute 1; H400 Aquatic Chronic 3; H412	>= 2,5 - < 10
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	63500-71-0 405-040-6 603-101-00-3 01-0000015458-64, 01-2119455547-30	Eye Irrit. 2; H319	>= 1 - < 10
vanillin	121-33-5 204-465-2 01-2119516040-60	Eye Irrit. 2; H319	>= 1 - < 10
1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one	33704-61-9 251-649-3 01-2119977131-40	Aquatic Chronic 2; H411 Skin Sens. 1B; H317 Eye Irrit. 2; H319 Skin Irrit. 2; H315	>= 2,5 - < 10
Propanedioic acid, 1-(3,3-dimethylcyclohexyl)ethyl ethyl ester	478695-70-4 455-260-1 01-0000019245-71	Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1	>= 2,5 - < 10
benzyl benzoate	120-51-4 204-402-9 607-085-00-9 01-2119976371-33	Acute Tox. 4; H302 Aquatic Chronic 2; H411 Aquatic Acute 1; H400	>= 2,5 - < 10
(Z)-3-hexenyl salicylate	65405-77-8 265-745-8 01-2119987320-37	Aquatic Chronic 2; H411 Aquatic Acute 1; H400 M-Factor (Acute	>= 1 - < 2,5

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		aquatic toxicity): 1	
[3aR-(3aα,5aβ,9aα,9bβ)]-dodecahydro-3a,6,6,9a-tetramethylnaphtho[2,1-b]furan	6790-58-5 229-861-2 01-2120070784-50-	Aquatic Chronic 3; H412	>= 1 - < 2,5
(10E)-1-oxacycloheptadec-10-en-2-one	63286-42-0 814-308-5 01-2120768557-38	Skin Sens. 1B; H317 Aquatic Acute 1; H400	>= 0,25 - < 1
(R)-p-mentha-1,8-diene	5989-27-5 227-813-5 601-029-00-7 01-2119529223-47	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Chronic 3; H412 Flam. Liq. 3; H226 Asp. Tox. 1; H304 Aquatic Acute 1; H400 M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
cinnamyl alcohol	104-54-1 203-212-3	Skin Sens. 1B; H317 Acute Tox. 4; H302 M-Factor (Acute aquatic toxicity): 1	>= 0,1 - < 1
ethyl (2E,4Z)-2,4-decadienoate	3025-30-7 221-178-8 01-2120769126-50	Skin Irrit. 2; H315 Aquatic Acute 1; H400 Aquatic Chronic 2; H411	>= 0,25 - < 1
2,6-di-tert-butyl-p-cresol	128-37-0 204-881-4 01-2119565113-46, 01-2119555270-46	Aquatic Chronic 1; H410 Aquatic Acute 1; H400	>= 0,25 - < 1
santalol	11031-45-1 234-262-4	Skin Sens. 1B; H317	>= 0,1 - < 1
7-hydroxycitronellal	107-75-5 203-518-7 01-2119973482-31	Eye Irrit. 2; H319 Skin Sens. 1B; H317 M-Factor (Acute aquatic toxicity): 1	>= 0,1 - < 1
(-)-pin-2(10)-ene	127-91-3 242-060-2 01-2119519230-54	Asp. Tox. 1; H304 Skin Irrit. 2; H315 Flam. Liq. 3; H226 Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1	>= 0,1 - < 0,25
p-mentha-1,4-diene	99-85-4 202-794-6 01-2120780478-40	Asp. Tox. 1; H304 Flam. Liq. 3; H226 Repr. 2; H361	>= 0,1 - < 0,25

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		Aquatic Chronic 2; H411	
isoeugenol	97-54-1 202-590-7 604-094-00-X	Acute Tox. 4; H312 Acute Tox. 4; H302 Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1A; H317 Acute Tox. 4; H332 STOT SE 3; H335 specific concentration limit Skin Sens. 1A; H317 ≥ 0,01 % Acute toxicity estimate Acute inhalation toxicity (dust/mist): 1,5 mg/l	≥ 0,01 - < 0,1
1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one	23696-85-7 245-833-2	Skin Sens. 1A; H317 Aquatic Chronic 2; H411 Skin Irrit. 2; H315 Acute toxicity estimate Acute dermal toxicity: 2.900 mg/kg	≥ 0,025 - < 0,1

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

- General advice : Take Hazard and Precautionary phrases (section 2) into account.
- If inhaled : Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice.
- In case of skin contact : Remove contaminated clothes. Wash thoroughly with water (and soap). Contact physician if symptoms persist.
- In case of eye contact : Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist.

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If swallowed : Rinse mouth with water and obtain medical advice.

4.2 Most important symptoms and effects, both acute and delayed

Risks : Causes skin irritation.
May cause an allergic skin reaction.
Causes serious eye irritation.

4.3 Indication of any immediate medical attention and special treatment needed

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Carbondioxide, dry chemical, foam.

Unsuitable extinguishing media : Do not use a direct waterjet on burning material.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-fighting : Water may be ineffective.

5.3 Advice for firefighters

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

Further information : Standard procedure for chemical fires.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a major spill.
Prevent spreading over a wide area (e.g. by containment or oil barriers).

6.2 Environmental precautions

Environmental precautions : Keep away from drains, surface- and groundwater and soil.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Clean up spillage promptly. Remove ignition sources. Provide adequate ventilation. Avoid excessive inhalation of vapours.
Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

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6.4 Reference to other sections

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use.

If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees.

Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.

Advice on protection against fire and explosion : Keep away from ignition sources and naked flame.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in a cool, dry, ventilated area away from heat sources. Keep containers upright and tightly closed when not in use.

7.3 Specific end use(s)

Specific use(s) : No information available.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

		Value type (Form of exposure)		Basis
2,6-di-tert-butyl-p-cresol	128-37-0	VME	10 mg/m3	FR VLE
Further information: Indicative exposure limits				

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

			Potential health effects	Value
1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-	Workers	Skin contact	Long-term local effects	0,1011 mg/cm2

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naphthyl)ethan-1-one				
Remarks:	Exposure time: 8 h			
	Workers	Skin contact	Long-term systemic effects	1,73 mg/kg bw/day
Remarks:	Exposure time: 8 h			
	Workers	Inhalation	Long-term systemic effects	1,76 mg/m3
Remarks:	Exposure time: 8 h			
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	Workers	Inhalation	Long-term systemic effects	44,1 mg/m3
Remarks:	REACH data			
	Workers	Dermal	Long-term systemic effects	41,7 mg/kg bw/day
Remarks:	REACH data			
	Workers	Eye contact	Local effects	
Remarks:	Available hazard data do not enable the derivation of a DNEL for eye irritant effects.			
	General population	Inhalation	Long-term systemic effects	13 mg/m3
Remarks:	REACH data			
	General population	Dermal	Long-term systemic effects	25
Remarks:	REACH data			
	General population	Oral	Long-term systemic effects	7,5 mg/kg bw/day
Remarks:	REACH data			

8.2 Exposure controls

Engineering measures

Where appropriate, use closed systems to transfer and process this material.
If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

Personal protective equipment

Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.
Equipment should conform to EN 166

Hand protection

Material : Nitrile rubber
Break through time : > 60 min
Glove thickness : 0,38 mm

Material : Nitrile rubber
Break through time : > 10 min
Glove thickness : 0,1 mm

Remarks : Avoid skin contact. Use chemically resistant gloves. The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived

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from it. Be aware that in daily use the durability of a chemical resistant protective glove can be notably shorter than the break through time measured according to EN 374, due to the numerous outside influences (e.g. temperature). Please observe the instructions regarding permeability and break-through time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time. The choice of an appropriate glove does not only depend on its material but also on other quality features and is different from one producer to the other.

Skin and body protection : Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.
Personal protection through wearing a tightly closed chemical protection suit and a self-contained breathing apparatus.

Respiratory protection : Use local exhaust ventilation around open tanks and other open sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures.
No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

- a) while engineering controls and appropriate safe work practices and/or procedures are being implemented; or
- b) during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or
- c) if normal operational workplace vapor concentration in the air is increased due to heat ;
- d) during emergencies; or
- e) if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.

Protective measures : To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material.
To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize

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exposure.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	: liquid
Colour	: orange to dark orange
Odour	: conforms to standard
Odour Threshold	: not determined
Melting point	: not determined
Boiling point	: not determined
Flammability	: not determined
Upper explosion limit / Upper flammability limit	: not determined
Lower explosion limit / Lower flammability limit	: not determined
Flash point	: 100,00 °C Method: closed cup
Auto-ignition temperature	: not determined
Decomposition temperature	: not determined
pH	: not determined
Viscosity, dynamic	: not determined
Viscosity, kinematic	: not determined
Water solubility	: not determined
Solubility in other solvents	: not determined
Partition coefficient: n-octanol/water	: not determined
Vapour pressure	: 0,1 hPa (20 °C) Calculated
Relative density	: 0,9445 - 0,9545
Density	: not determined

9.2 Other information

No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

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Hazardous reactions : Presents no significant reactivity hazard, by itself or in contact with water.

10.4 Conditions to avoid

Conditions to avoid : Direct sources of heat.

10.5 Incompatible materials

Materials to avoid : Avoid contact with strong acids, alkali or oxidizing agents.

10.6 Hazardous decomposition products

Carbon monoxide and unidentified organic compounds may be formed during combustion.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Not classified due to lack of data.

Product:

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Skin corrosion/irritation

Causes skin irritation.

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Species	: reconstructed human epidermis (RhE)
Assessment	: Causes skin irritation.
Method	: OECD Test Guideline 439
Result	: irritating
GLP	: yes
Test substance	: (undiluted)
Remarks	: REACH data

linalool:

Species	: Rabbit
Exposure time	: 4 h
Assessment	: Causes skin irritation.
Method	: OECD Test Guideline 404
Result	: Skin irritation
GLP	: yes
Test substance	: (undiluted)
Remarks	: REACH data

1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Species	: Human
Exposure time	: 0,25 h
Assessment	: Causes skin irritation.

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Method : EPISKIN Human Skin Model Test
Result : irritating
GLP : yes
Test substance : (undiluted)
Remarks : REACH data

Species : Human
Method : OECD Test Guideline 431
Result : No skin irritation
GLP : yes
Test substance : (undiluted)
Remarks : REACH data

Serious eye damage/eye irritation

Causes serious eye irritation.

Components:

linalool:

Species : Rabbit
Assessment : Causes serious eye irritation.
Method : OECD Test Guideline 405
Result : Eye irritation
GLP : no
Remarks : REACH data

tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans):

Species : Rabbit
Assessment : Causes serious eye irritation.
Method : Regulation (EC) No. 440/2008, Annex, B.5
Result : Eye irritation
GLP : yes
Remarks : REACH data

vanillin:

Remarks : No data available

Respiratory or skin sensitisation

Skin sensitisation

May cause an allergic skin reaction.

Respiratory sensitisation

Not classified due to lack of data.

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Test Type : Local lymph node assay (LLNA)
Species : Mouse
Assessment : The product is a skin sensitizer, sub-category 1B.
Method : OECD Test Guideline 429

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Result : Causes sensitisation.
GLP : yes

Remarks : REACH data

linalool:

Test Type : Local lymph node assay (LLNA)
Species : Mouse
Assessment : The product is a skin sensitiser, sub-category 1B.
Method : OECD Test Guideline 429
Result : The product is a skin sensitiser, sub-category 1B.
GLP : yes

Remarks : REACH data

1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Test Type : LLNA
Species : Mouse
Method : OECD 429
Result : The product is a skin sensitiser, sub-category 1B.
Remarks : REACH data

Test Type : repeated insult patch test
Species : Humans
Result : Does not cause skin sensitisation.
Test substance : 10.% in ethanol/DEP (75:25)
Remarks : REACH data

Species : Not tested on animals
Result : Does not cause respiratory sensitisation.
Remarks : REACH data

Germ cell mutagenicity

Not classified due to lack of data.

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Genotoxicity in vitro : Test Type: Microbial mutagenesis assay (Ames test)
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes
Remarks: REACH data

Test Type: Microbial mutagenesis assay (Ames test)
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

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Remarks: REACH data

Test Type: Chromosome aberration test in vitro
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: negative
GLP: yes
Remarks: REACH data

Test Type: In vitro mammalian cell gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes
Remarks: REACH data

Genotoxicity in vivo

: Test Type: In vivo micronucleus test
Species: Mouse (male and female)
Application Route: Dermal
Method: OECD Test Guideline 474
Result: negative
GLP: yes

Test Type: In vivo micronucleus test
Species: Rat (male and female)
Application Route: Dermal
Method: OECD Test Guideline 474
Result: negative
GLP: yes

linalool:

Genotoxicity in vitro

: Test Type: In vitro mammalian cell gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes
Remarks: REACH data

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: yes
Remarks: REACH data

Test Type: Microbial mutagenesis assay (Ames test)
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471

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Result: negative
GLP: yes
Remarks: REACH data

Genotoxicity in vivo : Test Type: In vivo micronucleus test
Species: Mouse (male and female)
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative
GLP: yes
Remarks: REACH data

1,4-dioxacyclohexadecane-5,16-dione:

Genotoxicity in vitro : Test Type: Microbial mutagenesis assay (Ames test)
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes
Remarks: REACH data

Test Type: Microbial mutagenesis assay (Ames test)
Test system: Escherichia coli
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes
Remarks: REACH data

Test Type: Chromosome aberration test in vitro
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: yes
Remarks: REACH data

Test Type: In vitro mammalian cell gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes
Remarks: Based on data from similar materials
REACH data

Carcinogenicity

Not classified due to lack of data.

Reproductive toxicity

Not classified due to lack of data.

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Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Species: Rat, male and female
Strain: wistar
Application Route: Oral
General Toxicity - Parent: NOAEL: ≥ 300 mg/kg body weight
Fertility: NOAEL: ≥ 300 mg/kg body weight
Method: OECD Test Guideline 443
GLP: yes
Remarks: REACH data

Test Type: reproductive and developmental toxicity study
Species: Rat, male and female
Strain: wistar
Application Route: Oral
General Toxicity - Parent: NOAEL: 120 mg/kg body weight
Fertility: NOAEL: ≥ 500 mg/kg body weight
Target Organs: Kidney, Liver, spleen
Method: OECD Test Guideline 421
GLP: yes
Remarks: REACH data

Effects on foetal development : Test Type: Pre-natal
Species: Rat, female
Application Route: Oral
Duration of Single Treatment: 21 d
Frequency of Treatment: 1 daily
General Toxicity Maternal: NOAEL: 240 mg/kg body weight
Developmental Toxicity: NOAEL: ≥ 480 mg/kg body weight
Method: OECD Test Guideline 414
GLP: yes
Remarks: REACH data

Test Type: Pre-natal
Species: Rabbit, female
Application Route: Oral
Duration of Single Treatment: 23 d
Frequency of Treatment: 1 daily
General Toxicity Maternal: NOAEL: 200 mg/kg body weight
Developmental Toxicity: NOAEL: ≥ 500 mg/kg body weight
Method: OECD Test Guideline 414
GLP: yes
Remarks: REACH data

linalool:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Species: Rat, male
Application Route: Oral
General Toxicity - Parent: NOAEL: 750 mg/kg body weight
General Toxicity F1: NOAEL: 200 mg/kg body weight
Method: OECD Test Guideline 421

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GLP: yes

Test Type: reproductive and developmental toxicity study
Species: Rat, female
Application Route: Oral
General Toxicity - Parent: NOAEL: 200 mg/kg body weight
General Toxicity F1: NOAEL: 200 mg/kg body weight
Method: OECD Test Guideline 421
GLP: yes

Effects on foetal development : Test Type: Pre-natal
Species: Rat, female
Application Route: Oral
Duration of Single Treatment: 11 d
Frequency of Treatment: 1 daily
General Toxicity Maternal: NOAEL: 500 mg/kg body weight
Developmental Toxicity: NOAEL: 1.000 mg/kg body weight
GLP: yes
Remarks: REACH data

1,4-dioxacyclohexadecane-5,16-dione:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Species: Rat, male
Strain: wistar
Application Route: Oral
General Toxicity - Parent: NOAEL: 300 mg/kg body weight
Fertility: NOAEL: 1.000 mg/kg body weight
Target Organs: Kidney
Method: OECD Test Guideline 421
GLP: yes
Remarks: REACH data

Test Type: reproductive and developmental toxicity study
Species: Rat, female
Strain: wistar
Application Route: Oral
General Toxicity - Parent: NOAEL: 1.000 mg/kg body weight
Fertility: NOAEL: 1.000 mg/kg body weight
Target Organs: Kidney
Method: OECD Test Guideline 421
GLP: yes
Remarks: REACH data

Effects on foetal development : Test Type: reproductive and developmental toxicity study
Species: Rat, male and female
Strain: wistar
Application Route: Oral
General Toxicity Maternal: NOAEL: 1.000 mg/kg body weight
Developmental Toxicity: NOAEL: 1.000 mg/kg body weight
Method: OECD Test Guideline 421
GLP: yes
Remarks: REACH data

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STOT - single exposure

Not classified due to lack of data.

STOT - repeated exposure

Not classified due to lack of data.

Repeated dose toxicity

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Species	: Rat, male and female
NOAEL	: 150 mg/kg
Application Route	: Oral
Exposure time	: 28-day
Number of exposures	: 1x /day
Method	: OECD Test Guideline 407
GLP	: yes
Remarks	: REACH data

Species	: Rat, male and female
NOAEL	: 120 mg/kg
Application Route	: Oral
Exposure time	: 90-day
Number of exposures	: 1x /day
Method	: OECD Test Guideline 408
GLP	: yes
Target Organs	: spleen, Liver, Kidney
Remarks	: REACH data

Species	: Rat, male and female
NOAEL	: 250 mg/kg
Application Route	: Dermal
Exposure time	: 90-day
Method	: OECD Test Guideline 411
GLP	: yes
Target Organs	: Skin, Liver
Remarks	: REACH data

Species	: Mouse, male and female
NOAEL	: 500 mg/kg
Application Route	: Dermal
Exposure time	: 90-day
Method	: OECD Test Guideline 411
GLP	: yes
Target Organs	: Skin, Liver
Remarks	: REACH data

linalool:

Species	: Rat, male
NOAEL	: >= 532,1 mg/kg
Application Route	: Ingestion
Exposure time	: 96 d
Method	: OECD Test Guideline 408

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Test substance	:	(undiluted)
GLP	:	yes
Remarks	:	REACH data
Species	:	Rat, female
NOAEL	:	>= 497,9 mg/kg
Application Route	:	Ingestion
Exposure time	:	95 d
Method	:	OECD Test Guideline 408
Test substance	:	(undiluted)
GLP	:	yes
Remarks	:	REACH data
Species	:	Rat, male and female
NOAEL	:	250 mg/kg
Application Route	:	Dermal
Exposure time	:	91 d
Method	:	OECD Test Guideline 411
Test substance	:	(undiluted)
GLP	:	yes
Remarks	:	REACH data

1,4-dioxacyclohexadecane-5,16-dione:

Species	:	Rat, male and female
NOAEL	:	1.000 mg/kg
Application Route	:	Oral
Exposure time	:	28-day
Number of exposures	:	1x /day
Method	:	OECD Test Guideline 407
GLP	:	yes
Remarks	:	REACH data

Species	:	Rat, male
NOAEL	:	300 mg/kg
Application Route	:	Oral
Method	:	OECD Test Guideline 421
GLP	:	yes
Target Organs	:	Kidney
Remarks	:	REACH data

Species	:	Rat, female
NOAEL	:	1.000 mg/kg
Application Route	:	Oral
Method	:	OECD Test Guideline 421
GLP	:	yes
Target Organs	:	Kidney
Remarks	:	REACH data

Aspiration toxicity

Not classified due to lack of data.

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11.2 Information on other hazards

Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 12: Ecological information

12.1 Toxicity

Components:

Propanedioic acid, 1-(3,3-dimethylcyclohexyl)ethyl ethyl ester:

M-Factor (Acute aquatic toxicity) : 1

12.2 Persistence and degradability

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge, non-adapted
Concentration: 18,8 mg/l
Result: Readily biodegradable.
Biodegradation: 96,3 %
Exposure time: 28 d
Method: OECD Test Guideline 301F
GLP: yes
Remarks: REACH data

linalool:

Biodegradability : Test Type: aerobic
Inoculum: see user defined free text
Concentration: 2 mg/l
Result: Readily biodegradable.
Biodegradation: 64,2 %
Exposure time: 28 d
Method: OECD Test Guideline 301D
GLP: yes
Remarks: REACH data

1,4-dioxacyclohexadecane-5,16-dione:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge, non-adapted
Concentration: 10 mg/l

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Result: Readily biodegradable.
Biodegradation: 100 %
Related to: Theoretical oxygen demand
Exposure time: 28 d
Method: OECD Test Guideline 301B
GLP: yes
Remarks: REACH data

12.3 Bioaccumulative potential

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Exposure time: 21 d
Bioconcentration factor (BCF): 391
Method: OECD Test Guideline 305
GLP: yes
Remarks: REACH data

linalool:

Bioaccumulation : Remarks: Accumulation in aquatic organisms is unlikely.

1,4-dioxacyclohexadecane-5,16-dione:

Bioaccumulation : Remarks: Does not bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 3,65 (20 °C)
Method: OECD Test Guideline 107
Remarks: REACH data

12.4 Mobility in soil

Product:

Mobility : Remarks: No data available

Components:

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Distribution among environmental compartments : Adsorption/Soil
Koc: 12598, log Koc: 4,1

1,4-dioxacyclohexadecane-5,16-dione:

Distribution among environmental compartments : Adsorption/Soil
Medium: Soil
Koc: 1800, log Koc: 3,3
Method: OECD 121

12.5 Results of PBT and vPvB assessment

Product:

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Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Product:

Additional ecological information : There is no data available for this product.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number or ID number

ADR : UN 3082

IMDG : UN 3082

IATA : UN 3082

14.2 UN proper shipping name

ADR : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(PROPANEDIOIC ACID, 1-(3,3-DIMETHYLCYCLOHEXYL)ETHYL ETHYL ESTER)

IMDG : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(PROPANEDIOIC ACID, 1-(3,3-DIMETHYLCYCLOHEXYL)ETHYL ETHYL ESTER)

IATA : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

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(PROPANEDIOIC ACID, 1-(3,3-DIMETHYLCYCLOHEXYL)ETHYL ETHYL ESTER)

14.3 Transport hazard class(es)

ADR	:	9
IMDG	:	9
IATA	:	9

14.4 Packing group

ADR	
Packing group	: III
Classification Code	: M6
Hazard Identification Number	: 90
Labels	: 9
Tunnel restriction code	: (-)
IMDG	
Packing group	: III
Labels	: 9
EmS Code	: F-A, S-F

IATA (Cargo)	
Packing instruction (cargo aircraft)	: 964
Packing instruction (LQ)	: Y964
Packing group	: III
Labels	: Miscellaneous Dangerous Goods

IATA_P (Passenger)	
Packing instruction (passenger aircraft)	: 964
Packing instruction (LQ)	: Y964
Packing group	: III
Labels	: Miscellaneous Dangerous Goods

14.5 Environmental hazards

ADR	
Environmentally hazardous	: yes

IMDG	
Marine pollutant	: yes (PROPANEDIOIC ACID, 1-(3,3-DIMETHYLCYCLOHEXYL)ETHYL ETHYL ESTER)

IATA (Passenger)	
Environmentally hazardous	: yes

IATA (Cargo)	
Environmentally hazardous	: yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

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14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles (Annex XVII) : Conditions of restriction for the following entries should be considered:
Number on list 75, 3

phenylacetonitrile
phenol
4-allylveratrole
4-allylanisole
methyl oct-2-ynoate
p-mentha-1,3-diene
trans-hex-2-enal
benzyl salicylate
cinnamaldehyde
geraniol
citral
methyl salicylate
1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one
isoeugenol
eugenol
(-)-pin-2(10)-ene
7-hydroxycitronellal
benzoic acid
cinnamyl alcohol
(R)-p-mentha-1,8-diene
vanillin
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors : acetone

REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59). : Not applicable

Regulation (EC) No 1005/2009 on substances that deplete the ozone layer : Not applicable

Regulation (EU) 2019/1021 on persistent organic pollutants (recast) : Not applicable

REACH - List of substances subject to authorisation (Annex XIV) : Not applicable

Regulation (EU) 2019/1148 on the marketing and use of explo-

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sives precursors

acetone (ANNEX II)

Reinforced medical supervision (R4624-23) : The product has no CMR properties category 1, 1A or 1B

Other regulations:

Take note of Directive 94/33/EC on the protection of young people at work or stricter national regulations, where applicable.

15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

SECTION 16: Other information

Full text of H-Statements

H226	: Flammable liquid and vapour.
H302	: Harmful if swallowed.
H304	: May be fatal if swallowed and enters airways.
H312	: Harmful in contact with skin.
H315	: Causes skin irritation.
H317	: May cause an allergic skin reaction.
H319	: Causes serious eye irritation.
H332	: Harmful if inhaled.
H335	: May cause respiratory irritation.
H361	: Suspected of damaging fertility or the unborn child.
H400	: Very toxic to aquatic life.
H410	: Very toxic to aquatic life with long lasting effects.
H411	: Toxic to aquatic life with long lasting effects.
H412	: Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Acute	: Short-term (acute) aquatic hazard
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Asp. Tox.	: Aspiration hazard
Eye Irrit.	: Eye irritation
Flam. Liq.	: Flammable liquids
Repr.	: Reproductive toxicity
Skin Irrit.	: Skin irritation
Skin Sens.	: Skin sensitisation
STOT SE	: Specific target organ toxicity - single exposure
FR VLE	: France. Occupational Exposure Limits
FR VLE / VME	: Time Weighted Average

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regula-

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tion (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information : In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110].
In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace".
Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.
The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration $\geq 1.0\%$ due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the

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workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

FR / EN